

Scenario Analysis

METIMMOX-1 Economic Evaluation

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2026-06-19

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1 Overview

This report summarizes scenario EVPPI and PSA outputs for the single joint economic model. The expected cache is `scenario_evppi_results_{utility}.rds`, with a preview cache used as a fallback if available.

2 Scenario Definitions

Table 1: Scenario definitions loaded from `scenario_evppi_results_correct.rds`

scenario_id	scenario_name	wtp	c_drug_nivo
base	Base Case	EUR 51,000	EUR 13,923
s1_decreased_nivo	Decreased Nivolumab Cost	EUR 51,000	EUR 4,641
s2_wtp100k	WTP €100,000	EUR 100,000	EUR 13,923
s3_wtp100k_decreased	WTP €100,000 + Decreased Nivolumab	EUR 100,000	EUR 4,641
s4_wtp150k	WTP €150,000	EUR 150,000	EUR 13,923
s5_wtp150k_decreased	WTP €150,000 + Decreased Nivolumab	EUR 150,000	EUR 4,641

3 PSA Summary by Scenario

Table 2: PSA summary by scenario

	Scenario	WTP	Strategy	Mean Cost	Mean QALYs	Mean NMB
control...1	Base Case	EUR 51,000	Standard of Care	EUR 21,156	1.307	EUR 45,493
crp...2	Base Case	EUR 51,000	CRP-guided	EUR 55,173	1.411	EUR 16,793
tmb_braf...3	Base Case	EUR 51,000	TMB/BRAF-guided	EUR 60,943	1.394	EUR 10,156
control...4	WTP €100,000	EUR 100,000	Standard of Care	EUR 21,156	1.307	EUR 109,529
crp...5	WTP €100,000	EUR 100,000	CRP-guided	EUR 55,173	1.411	EUR 85,938
tmb_braf...6	WTP €100,000	EUR 100,000	TMB/BRAF-guided	EUR 60,943	1.394	EUR 78,468
control...7	WTP €150,000	EUR 150,000	Standard of Care	EUR 21,156	1.307	EUR 174,872
crp...8	WTP €150,000	EUR 150,000	CRP-guided	EUR 55,173	1.411	EUR 156,493
tmb_braf...9	WTP €150,000	EUR 150,000	TMB/BRAF-guided	EUR 60,943	1.394	EUR 148,173
control...10	Decreased Nivolumab Cost	EUR 51,000	Standard of Care	EUR 21,178	1.305	EUR 45,365
crp...11	Decreased Nivolumab Cost	EUR 51,000	CRP-guided	EUR 32,295	1.410	EUR 39,639
tmb_braf...12	Decreased Nivolumab Cost	EUR 51,000	TMB/BRAF-guided	EUR 35,195	1.392	EUR 35,797
control...13	WTP €100,000 + Decreased Nivolumab	EUR 100,000	Standard of Care	EUR 21,178	1.305	EUR 109,298
crp...14	WTP €100,000 + Decreased Nivolumab	EUR 100,000	CRP-guided	EUR 32,295	1.410	EUR 108,752
tmb_braf...15	WTP €100,000 + Decreased Nivolumab	EUR 100,000	TMB/BRAF-guided	EUR 35,195	1.392	EUR 104,006
control...16	WTP €150,000 + Decreased Nivolumab	EUR 150,000	Standard of Care	EUR 21,178	1.305	EUR 174,535
crp...17	WTP €150,000 + Decreased Nivolumab	EUR 150,000	CRP-guided	EUR 32,295	1.410	EUR 179,276
tmb_braf...18	WTP €150,000 + Decreased Nivolumab	EUR 150,000	TMB/BRAF-guided	EUR 35,195	1.392	EUR 173,607

4 Scenario EVPPI

Table 3: Top EVPPI parameters by scenario

Scenario	WTP	Parameter	EVPPI	Percent of EVPI	Population EVPPI
Base Case	EUR 51,000	u_np	EUR 208	62.6%	EUR 2.59M
Base Case	EUR 51,000	[GROUP] test_costs	EUR 152	45.9%	EUR 1.90M
Base Case	EUR 51,000	u_p	EUR 127	38.2%	EUR 1.58M
Base Case	EUR 51,000	c_test_CT	EUR 88	26.6%	EUR 1.10M
Base Case	EUR 51,000	c_other_visit	EUR 43	13.1%	EUR 0.54M
Decreased Nivolumab Cost	EUR 51,000	[GROUP] utilities	EUR 465	10.8%	EUR 5.80M
Decreased Nivolumab Cost	EUR 51,000	[GROUP] other_costs	EUR 260	6.0%	EUR 3.24M
Decreased Nivolumab Cost	EUR 51,000	[GROUP] all_costs	EUR 239	5.5%	EUR 2.98M
Decreased Nivolumab Cost	EUR 51,000	u_np	EUR 187	4.3%	EUR 2.33M
Decreased Nivolumab Cost	EUR 51,000	[GROUP] test_costs	EUR 146	3.4%	EUR 1.82M
WTP €100,000	EUR 100,000	u_np	EUR 421	9.3%	EUR 5.25M
WTP €100,000	EUR 100,000	u_p	EUR 235	5.2%	EUR 2.93M
WTP €100,000	EUR 100,000	[GROUP] test_costs	EUR 202	4.5%	EUR 2.53M
WTP €100,000	EUR 100,000	c_test_CT	EUR 167	3.7%	EUR 2.09M
WTP €100,000	EUR 100,000	c_test_blood	EUR 109	2.4%	EUR 1.36M
WTP €100,000 + Decreased Nivolumab	EUR 100,000	[GROUP] utilities	EUR 1,992	13.5%	EUR 24.85M
WTP €100,000 + Decreased Nivolumab	EUR 100,000	u_np	EUR 1,222	8.3%	EUR 15.24M
WTP €100,000 + Decreased Nivolumab	EUR 100,000	c_drug_nivo	EUR 478	3.2%	EUR 5.96M
WTP €100,000 + Decreased Nivolumab	EUR 100,000	u_p	EUR 477	3.2%	EUR 5.95M
WTP €100,000 + Decreased Nivolumab	EUR 100,000	[GROUP] drug_costs	EUR 476	3.2%	EUR 5.93M
WTP €150,000	EUR 150,000	u_np	EUR 639	5.0%	EUR 7.97M
WTP €150,000	EUR 150,000	u_p	EUR 345	2.7%	EUR 4.31M
WTP €150,000	EUR 150,000	[GROUP] test_costs	EUR 315	2.5%	EUR 3.93M
WTP €150,000	EUR 150,000	c_test_CT	EUR 248	2.0%	EUR 3.10M
WTP €150,000	EUR 150,000	c_test_blood	EUR 177	1.4%	EUR 2.20M
WTP €150,000 + Decreased Nivolumab	EUR 150,000	u_np	EUR 887	4.1%	EUR 11.06M
WTP €150,000 + Decreased Nivolumab	EUR 150,000	[GROUP] utilities	EUR 819	3.8%	EUR 10.22M
WTP €150,000 + Decreased Nivolumab	EUR 150,000	p_crp	EUR 198	0.9%	EUR 2.47M
WTP €150,000 + Decreased Nivolumab	EUR 150,000	u_p	EUR 157	0.7%	EUR 1.96M
WTP €150,000 + Decreased Nivolumab	EUR 150,000	c_test_blood	EUR 144	0.7%	EUR 1.80M

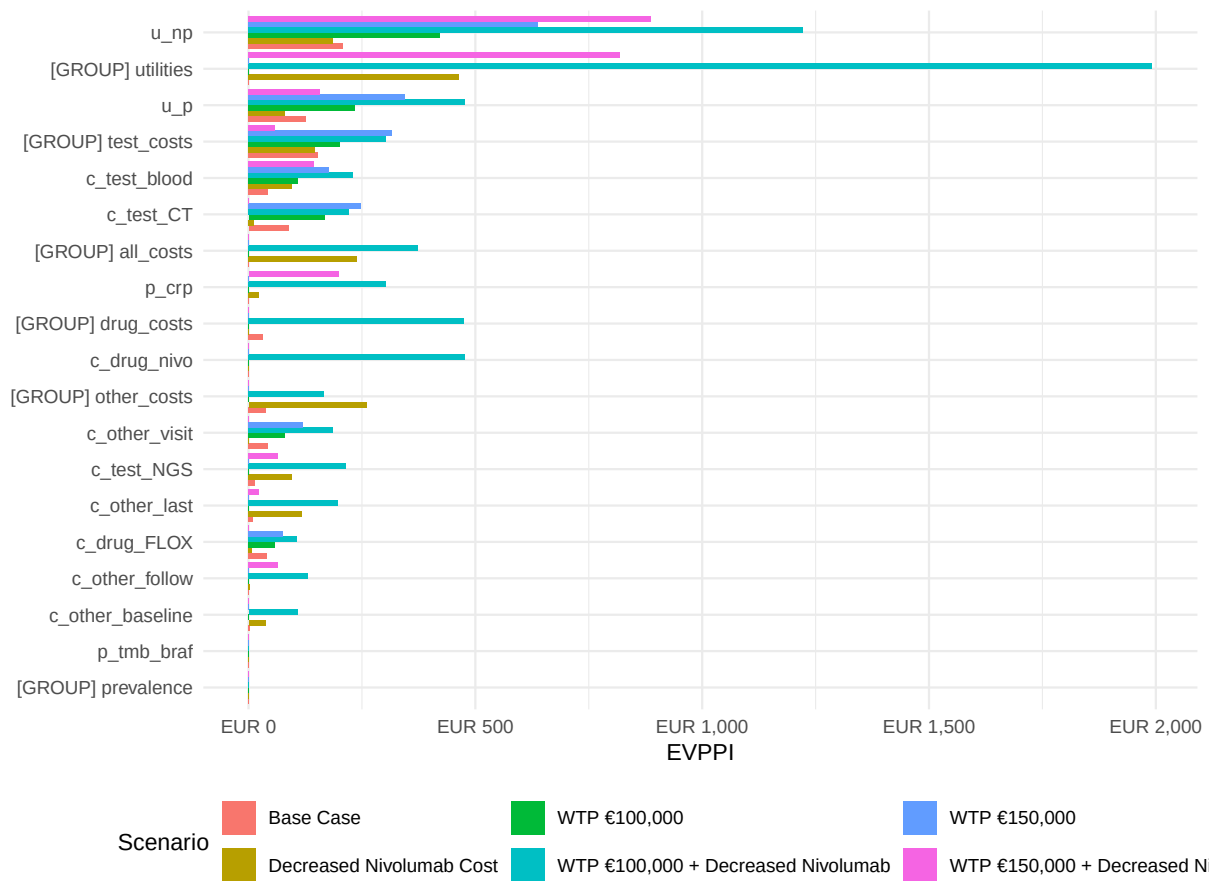


Figure 1: Scenario EVPPI by parameter

Report completed on: 2026-06-19
Repository: ben-geisler/METIMMOX-1
Report version: 4.0